

Cohort Analysis Report on GA4 style Dataset

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Business Problem Statement:

Which customer acquisition cohorts generate the highest long-term value?

Which marketing channels acquire customers who actually return?

Which seasonal campaigns create lasting customers rather than one-time buyers?

Executive Summary:

Main business takeaway:

Tab 1 highlights that among the channel-based cohorts, organic search, organic paid channels have acquired most customers with highest revenue generated individually. Organic search and Direct channel also show better D30 retention. **There is evidence for strong Product-Market fit with overall D30 bounded retention at 10.66% (industry standards).**

In terms of campaign-based cohort, Meta retargeting and always on search campaigns are generating highest Average order value. Push and meta retargeting are showing better retention rates, suggesting acquisition of quality customers over google shopping where repeat purchase rate is higher but revenue is low. The customers from the former campaigns can be seasonal or discount buyers who has longer purchase cycle and latter ones are recurring consumables buyers.

Monthly cohorts show an average retention at 4% only out of an average 34% acquired at the beginning. Around 4/5th of the devices used for shopping is mobile devices.

Tab 2 highlights among the new customers only 38% has made it to loyal segment and 39% have become dormant. Around 36% have made second purchase but only a mere 6% has made 3, 4+ purchase in the time period of 3-6 months.

Highest LTV is coming from “Always on search” campaign for D30 and “Instagram influencers” for D90.

Tab 3 highlights in the funnel steps, worst leakage is happening in product view to add to cart step (only 21% added items in the cart). As for the engagement depth, on average less than 1 product is viewed in a session. **This hints towards large homepage bounces.** While for repeat purchases, Direct and Affiliate channels are doing well.

In the RFM segment, around 1.18k are in the Can't Lose category, meaning they used to buy regularly (in the D90, D180 LTV milestone) but has stopped returning (low RFM scores). 2.48k customers are already in the Lost category.

Women segment is leading the product category followed by Men category.

Top Marketing KPIs:

Total acquired users – **3.912k** | Best channel return rate (Push) – **12.18%**

Avg. revenue LTV – **\$9.79k** | Avg. order per user – **3.17**

Avg. D90 retention – **17.05%** | Avg. D30 retention – **10.66%**

Product view rate% - **75.62%** | Cart addition rate% - **16.09%**

Cart to purchase rate% - **37.64%**

Methodology and Data Sources:

Methodology: The analysis is done using Power BI wherein the 4 tabs in the dashboard analyze customer data in multiple cohorts namely channel-based, campaign-based, time-based and based on their lifetime value. The core metrics include

- Channel & Campaign-level acquisition comparison,
- D7-D30-D60 retention matrix,
- Channel-level revenue comparison,
- LTV by acquisition month,
- Customer lifecycle funnel leakage,
- Purchase frequency,
- Engagement vs Repeat purchase and
- RFM segment distribution.

Source: The data is a static excel file and a replica of GA4 style data. It is generated using python code. The given results do not resemble or suggest the actual market trends.

Key findings and trends:

Tab 1: Acquisition cohorts:

1. New users by acquisition channel

Which channel is driving traffic surge:

Highest traffic from Organic search at (311+300+348) customers, followed by organic social and paid search

2. Channel cohort retention grid

Which channel’s users actually come back:

Organic search, Organic social and Paid search are top 3 in acquisition at 24.51%, 18.40% and 15.49% but Organic search and Direct channels are retaining better at 2.81% and at 1.41% on D30 than organic social and paid search. Therefore, showing a pattern of “Efficiency leak”.

Retention by Acquisition Channel					
channel_group		D1	D7	D30	D60
Affiliate	7.00%	0.74%	0.38%	0.33%	0.31%
Direct	10.07%	1.66%	1.38%	1.41%	0.64%
Email	7.69%	1.43%	1.28%	1.18%	0.87%
Organic Search	24.51%	4.58%	3.63%	2.81%	1.92%
Organic Social	18.40%	1.84%	1.33%	1.25%	0.64%
Paid Search	15.49%	2.22%	1.87%	1.38%	1.05%
Paid Social	11.78%	1.48%	1.53%	1.02%	0.64%
Push	5.04%	1.12%	0.84%	0.69%	0.64%

Product-market fit:

Under 3% D30 signals major onboarding friction, technical bugs, or irrelevant notification spamming. 5-7% shows standard industry performance; requires consistent paid re-engagement to sustain. And 8-14% shows strong product-market fit, excellent habit loops, and relevant messaging.

Here, the D30 bounded retention rate for all channels is 10.66% which shows strong product-market fit. It confirms long-term "stickiness". Users active at this point have officially integrated the ecommerce app into their routine.

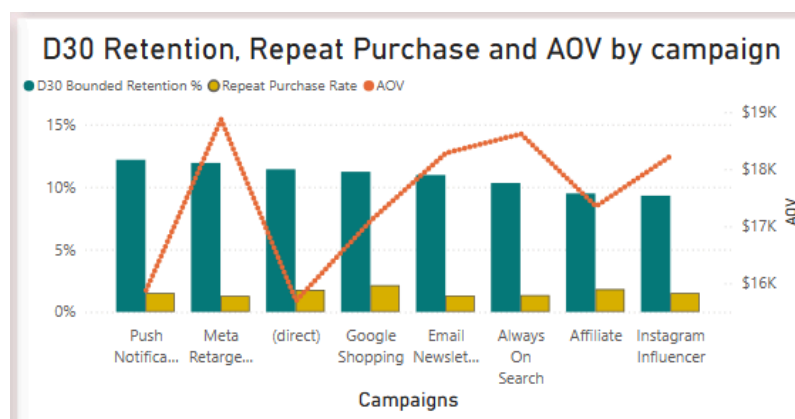
3. Campaign cohort comparison

Which campaigns create lasting customers:

Meta retargeting campaign customers have highest average order value (AOV), around \$18.8k, followed by Always on search at \$18.6k. Highest D30 bounded retention is from Push notifications at 12.18% followed by Meta Retargeting at 11.91%.

But highest repeat purchases are from Google shopping at 2.15% followed by Affiliate at 1.82%. This combination indicates following consumer behavior pattern:

- There are large number of consumers buying huge quantities of products during a seasonal sale such as EOSS and they are not required to buy any sooner.
- High D30 retention may suggest that consumers from top campaigns are engaging with the app by opening the app and viewing products but they are not buying anything.
- The **consumers from top AOV and top D30 retention campaigns can be loyal but their natural purchasing cycle might be longer** (Avg. D90 retention is 17.05%, higher than D30 retention).



Moving forward, it is required that we analyze their product-mix (if they are buying long-lasting products), adjust re-targeting timelines and monetize engagement by including non-purchase options such as referral.

Simultaneously, google and direct, excel at driving continuous and habitual buying behavior indicating strong loyalty. The consumers are not high-value therefore, they might be promoting low-friction, recurring consumable products. They browse more deeply and add to cart more intentionally. Their long-term compounded LTV can create stable and predictable revenue stream.

4. Monthly cohort acquisition trend

New cohorts are not retaining any better. The slowdown problem is getting worse. From an average of 31-36% of acquisition at the beginning of Jan-Mar, only an average of 4% is retained between Apr-June.

5. Acquisition channel revenue share

What % of total revenue comes from each channel's customers:

Highest revenue is coming from organic search at \$2.7mn (24.75%), followed by organic social and paid search at \$2.13mn (18.91%) and \$1.73mn (15.31%) respectively. As we have already noted, these channels are acquiring a large number of consumers, therefore, their behavioral patterns might be similar to those discussed in point 3.

6. Device x channel acquisition breakdown

Device mix within each channel:

70-78% of sessions took place in mobile phones for every channel, followed by desktop at 16-25%. This reflects a typical ecommerce behavior where the consumers value convenience, immediacy and accessibility.

Tab 2: Retention / LTV

1. Monthly cohort retention heatmap

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2. LTV curve by acquisition cohort

Cumulative ARPU (average revenue per user) is better for Jan Cohort at \$3039 followed by March Cohort. Average revenue per user (LTV) is \$1.63k.

3. Customer lifecycle stage distribution

User progression through the lifecycle:

Around 380 dormant customers shows that these are not returning frequently enough. A smaller number of customers, only 370 is loyal for the platform as compared to the 980 new ones acquired.

Stage distribution over time:

Each lifecycle stage is almost equally distributed monthly with a slightly bigger difference in March new customer acquisition numbers. On an average, every month, 65 such customers were acquired who are at the risk of churning now (total 196 for 3 months).

4. Purchase frequency distribution

Out of total 3.9k acquired customers and 810 active consumers, only 290 consumers have made 2nd purchase in a span of 3 months. A mere 40 and 10 consumers have made 3, 4+ purchases.

This suggests that the acquired customers (active) just used the initial welcome code for making a purchase and now engage with the platform for only viewing products. Alternatively, have longer purchasing cycles.

5. Seasonal cohort LTV comparison

For D30 and D90 cohorts, "Always on search" campaigns brought highest LTV at \$109k and \$264k. For D180, "Instagram influencer" was leader with \$120k, reinforcing the fact that these consumers only wait for good deals to make a purchase (any festival, season) and come back after a longer period for another such sale.

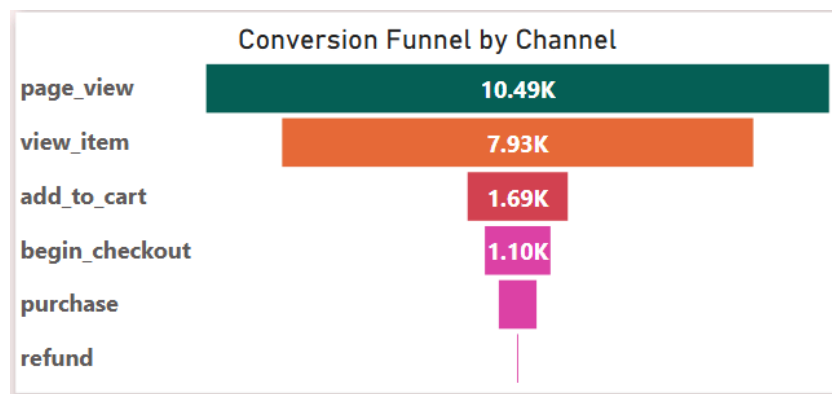
Lowest LTV is coming from Push notifications and Affiliate campaigns reinforcing that they are our low-friction, recurring consumable buyers whose value compounds over time.

Tab 3: Behavioral analysis:

1. Conversion funnel by channel

Which funnel step leaks worst for each channel:

There is a massive difference between product view (7.93k) and add to cart (1.69k) metrics which suggests about the quality of consumers acquired. It means either the right audience is not being targeted or there is a high mismatch between the campaign promise and the quality (design/trend/type) of actual products offered. Because even from those adding products to cart, less than 50% (635) are actually proceeding to purchase products.



Additionally, this may include the consumers who have longer purchasing cycles due to bundle purchasing in seasonal sales and they view products to look for deals / compare their prices on competitor apps.

Where to intervene:

The focus should be on targeting and acquiring quality customers and to understand their deeper needs. Further analysis can be done on product features used, scraping online comments to gauge their platform experience and reviews or asking them to fill a direct feedback survey to understand whether there is any friction in using this platform or are they just unable to find the usefulness of the platform in their life.

The funnel proportions are almost same for all the acquisition channels.

2. Engagement depth vs repeat purchase rate

Whether deeper engagement predicts retention:

On average, less than 1 product is viewed per session, suggesting high landing page bounce rate due to misleading traffic from campaigns, slow load times or not finding the page indulging or needed enough to explore any products (UI/UX issue).

Among those, beyond the 0.75 product view per user (overall average) threshold, email, direct and affiliate acquisition channels have few customers with 4-5% of repeat purchase rate (RPR). Otherwise, mostly fall within 1-2% of repeat rate.

At what engagement threshold repeat-buy probability inflects:

For Jan Cohort, at 75.84% Product view rate, 132 customers showed a RPR of 4.55% (Direct).

For March Cohort, at 80.97% PVR, 90 customers showed a RPR of 4.44% (Affiliate).

To counter this issue check page load speed, analyze user session recordings with tools like Hotjar and make your best-selling products visible on top of the page.

3. RFM segment distribution

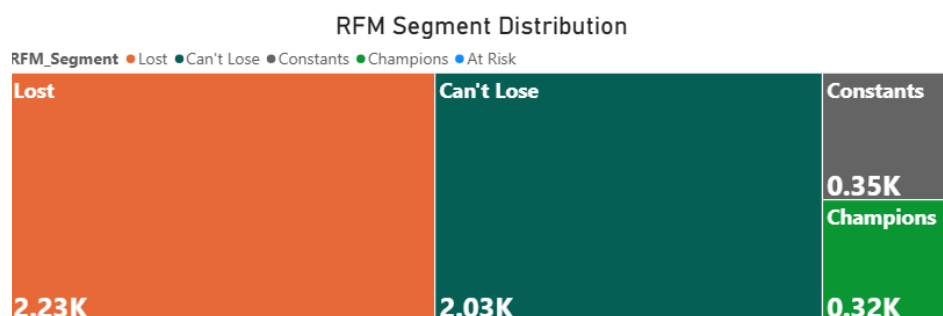
Core retention problem – At risk vs Champions ratio:

At risk (9) customers are really low as compared to Champions (204) but interestingly a great number of consumers are already “Lost” (2480) i.e. those who are not interacting with platform at all. Also, a large number of consumers are in the segment of “Can’t lose” (1188) i.e. those who were active and loyal at one point but has stopped purchasing now. Therefore, further action plans should take this segment “Can’t Lose” into focus and try to re-engage them through offers, discounts, feedback surveys and resolving any pending disputes regarding purchases or experience.

In user drill through, we can see that these customers belong to D90 and D180 LTV Milestone but have an average RFM score between 1.67 to 2.33 (out of 5). This highlights that they are quality customers of the platform but losing engagement due to unknown factors (platform friction, quality issues, new competition in

the market or having a much longer purchase cycle). Therefore, a feedback survey becomes important for such consumers to thoroughly understand their needs and see if they are actually receiving value from the platform and will purchase in future or they might churn soon due to issues given above.

For “At Risk” and “Lost” consumer groups, prepare a suitable churn reduction strategy. A survey becomes equally important for them as well to understand more about your product-market fit.



4. Customer dimension distribution

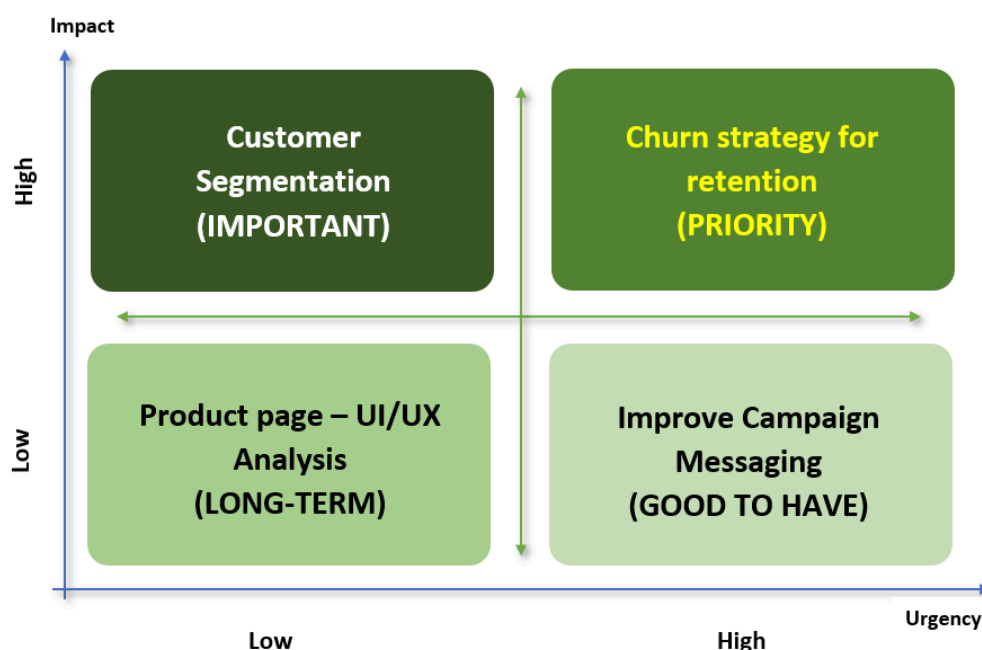
Within the product category, Women category is most sold at 38.44% followed by Men category 26.87%. Kids section is the least sold at 4.56% suggesting this category is still evolving in online domain or becoming a specialized category where people prefer dedicated platforms to buy kids clothes.

Affiliate, Email and Push channels are selling more women segment products at 48.84%, 47.76% and 46.88%. Paid Search and Paid Social channels are bringing almost equivalent Women and Men segment buyers.

Our Direct channel cohort customers are majorly making purchases on Thursday (16.28%) followed by Saturday (15.41%). In that, women segment is selling the most, via mobile. For rest of the channels, the data is almost equally divided (due to data being synthetic and not actual market data).

In women segment, Delhi people are buying more on Tuesdays and Fridays while Chennai people are buying more on Tuesdays and Sundays. In Jaipur, Saturdays and Thursdays. In Pune, mostly Sundays (around 35%)

Actionable recommendations:



Churn strategy:

A large pool of acquired customers has already been “Lost”. Assuming that was an error of not targeting the right audience, you will work on identifying right target audience by customer segmentation. But apart from them, a large ratio of users is on the verge of abandoning the app or already done so even if they are viewing products here and there.

Therefore, **prioritize “Can’t lose” and “At Risk” segment by thoroughly understanding their needs and re-engage them with platform by bringing usefulness to them. Because retaining existing customers is more cost-efficient than gaining new ones.** Hence, it is really important to understand whether these people actually have a longer purchase cycle or we have lost them to the competition. First, through an exit survey understand why customers churn, either due to technical friction, poor onboarding or price shock. Secondly, intervene in their non-purchasing session through loyalty building tactics (loyalty incentives, offer codes, product walkthroughs etc.).

Also, time is of essence here, for customers with longer purchasing cycle, do not spam them but for those leaving for competition, immediately apply retargeting tactics.

Customer segmentation:

Out of the newly acquired customers, a very small % of them converted into or made into loyal segment. A similar amount of them have become dormant. Creating their detailed persona will help you to identify in advance why they make it to a particular segment. Their purchase frequency, engagement and CLV / retention should be analyzed for a much longer period (e.g. 9-12 months instead of just 3 months) and the persona traits arrived at should be mapped against the motivations derived from feedback surveys or interviews.

Because before creating or improving a platform or forming brand messaging, it is important to identify the consumers who are more likely to respond to those messages. Accordingly, budget can be allocated to the right segment. An initial idea about segments has been set-up in this project through RFM Analysis based on their real purchasing behaviors rather than assumptions. Take these segments as a base and examine market trends, competitor’s offering and new channels to market yourself. This research will help you to focus your customer segmentation study towards **acquiring right target audience apart from retention (priority) to specifically deal with high landing page bounce rates at the top of the funnel.**

Here, we already have a Value-based segmentation on our consumers. In next steps, you must support it with Psychographic segmentation (to improve brand-tone) and Needs-based segmentation (to align product features with customer needs). To further improve engagement, Behavioral segmentation (check patterns and predict) can also be done.

Improve campaign messaging:

To **counter high page bounce rates**, measuring the quality of customers acquired via campaigns is necessary. A misleading traffic might occur due to misleading campaign messaging. It is required to understand whether there was a promise mismatch, price issue for the people who clicked on the ad or they already had low intentions of buying the product and clicked by accident.

For such cases, an exit survey (2-3 questions) should be added in the platform to assess the quality of customers attracted via campaigns. Additionally, a detailed long-form survey or in-depth interview can be conducted with some of those lapsed to gain deeper understanding of their needs and expectations from an e-commerce platform. Aide this with competitive benchmarking and social listening.

Product page analysis and improve UI/UX:

A massive gap between product views and add-to-cart actions indicates that while the marketing successfully gets users interested enough to look at your items, **the product page itself fails to convince them to buy.** The reasons may vary from hidden costs, lack of reviews to lack of good CTA buttons or the product size being out of stock. Therefore, improve product page outlook by adding introductory discounts, visible CTA buttons and risk-free return policies for building trust in absence of reviews. Further:

- Check if there are any page load issues (e.g. landing page takes more than 2 seconds in loading) during heavy traffic.
- Analyze user session recordings to mark any friction in using the app and observing at which funnel step the consumers are abandoning the platform majorly.
- Keep your best-sellers on the top along with other products based on the combinations of product type x brand preferred by a particular customer segment.
- Counter any other design issues which may occur in survey results.